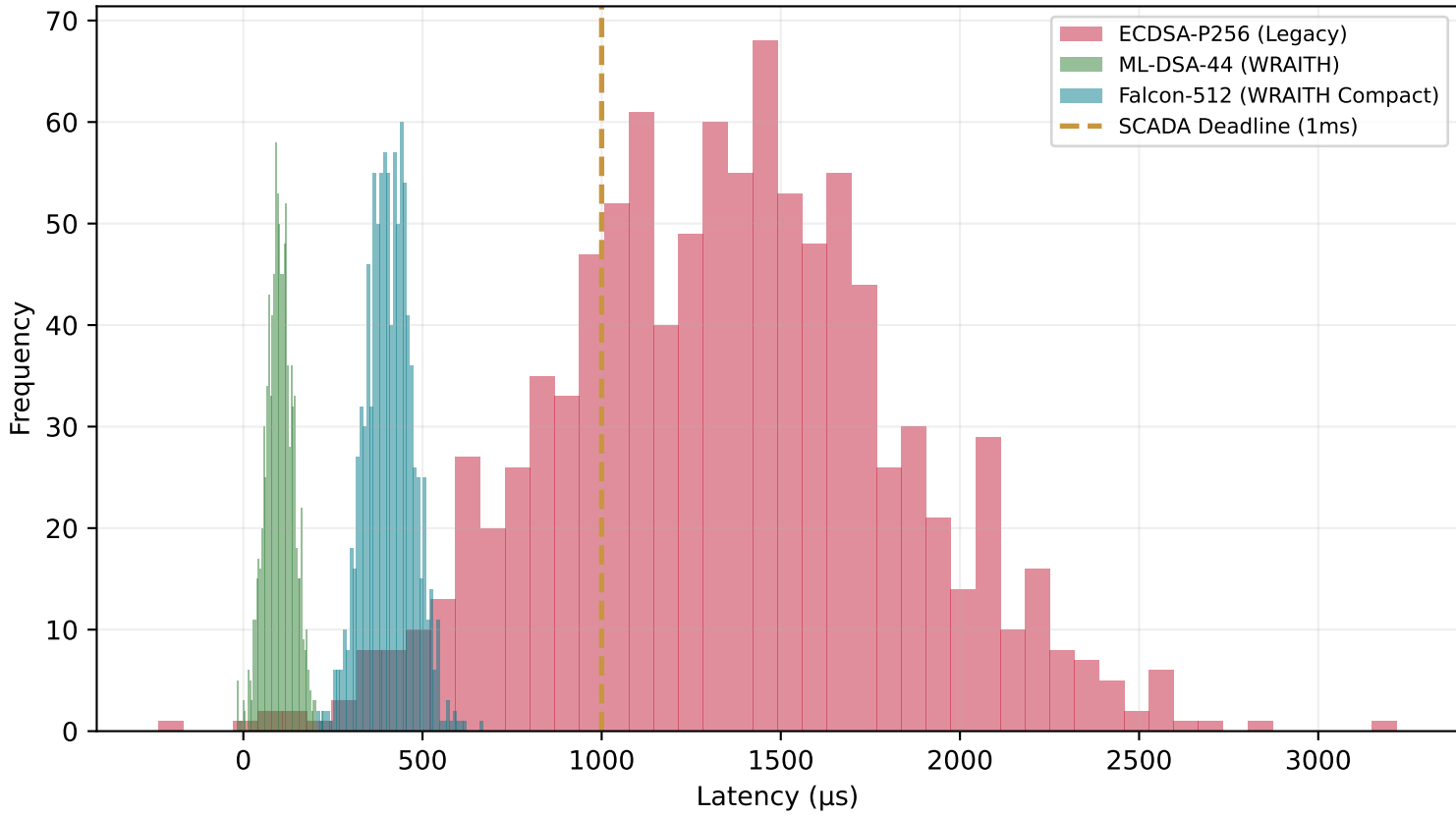


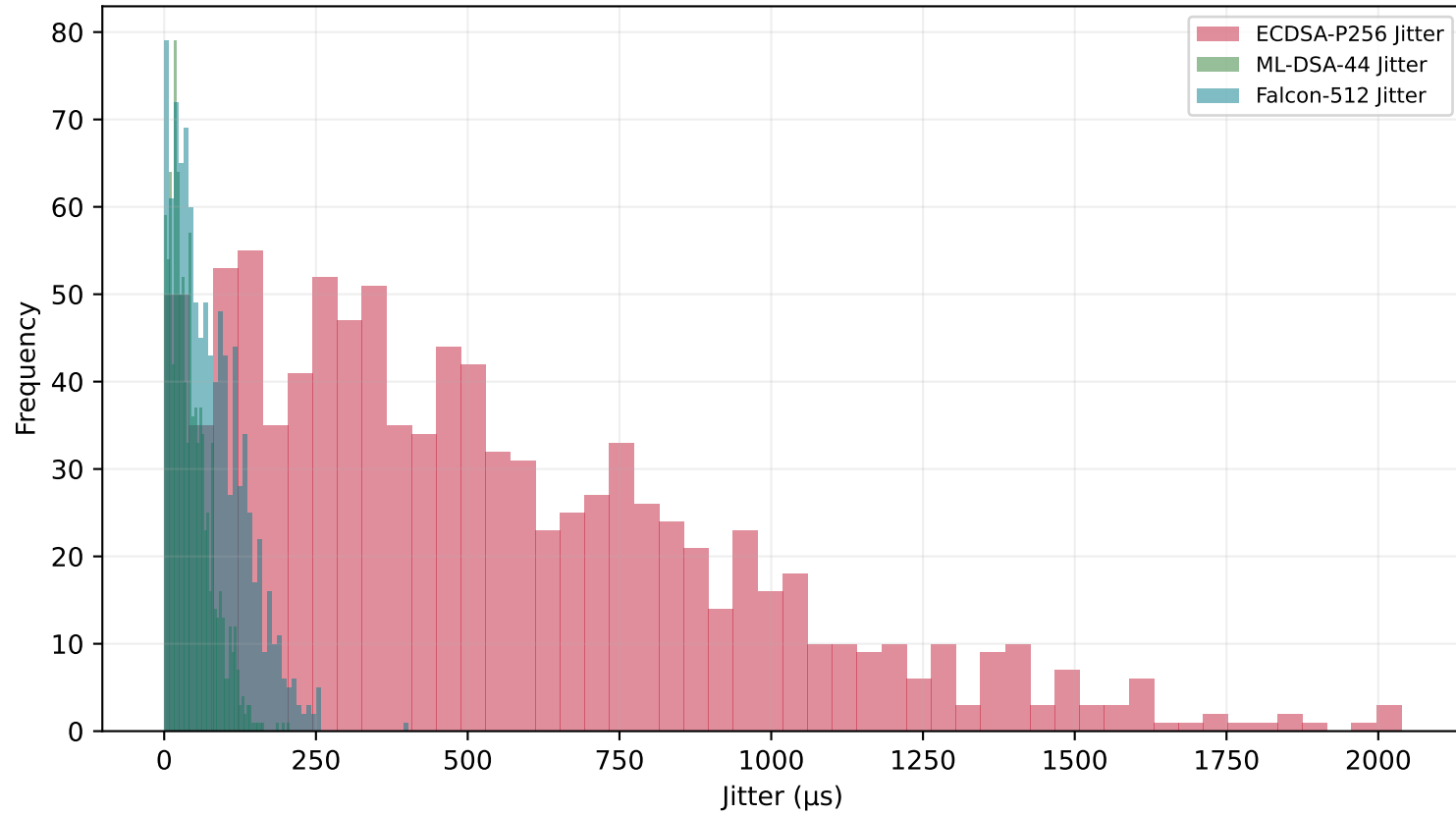
SCADA-Ready PQC Latency & Jitter Analysis

Gulf Infrastructure — Modbus/DNP3 Real-Time Constraints

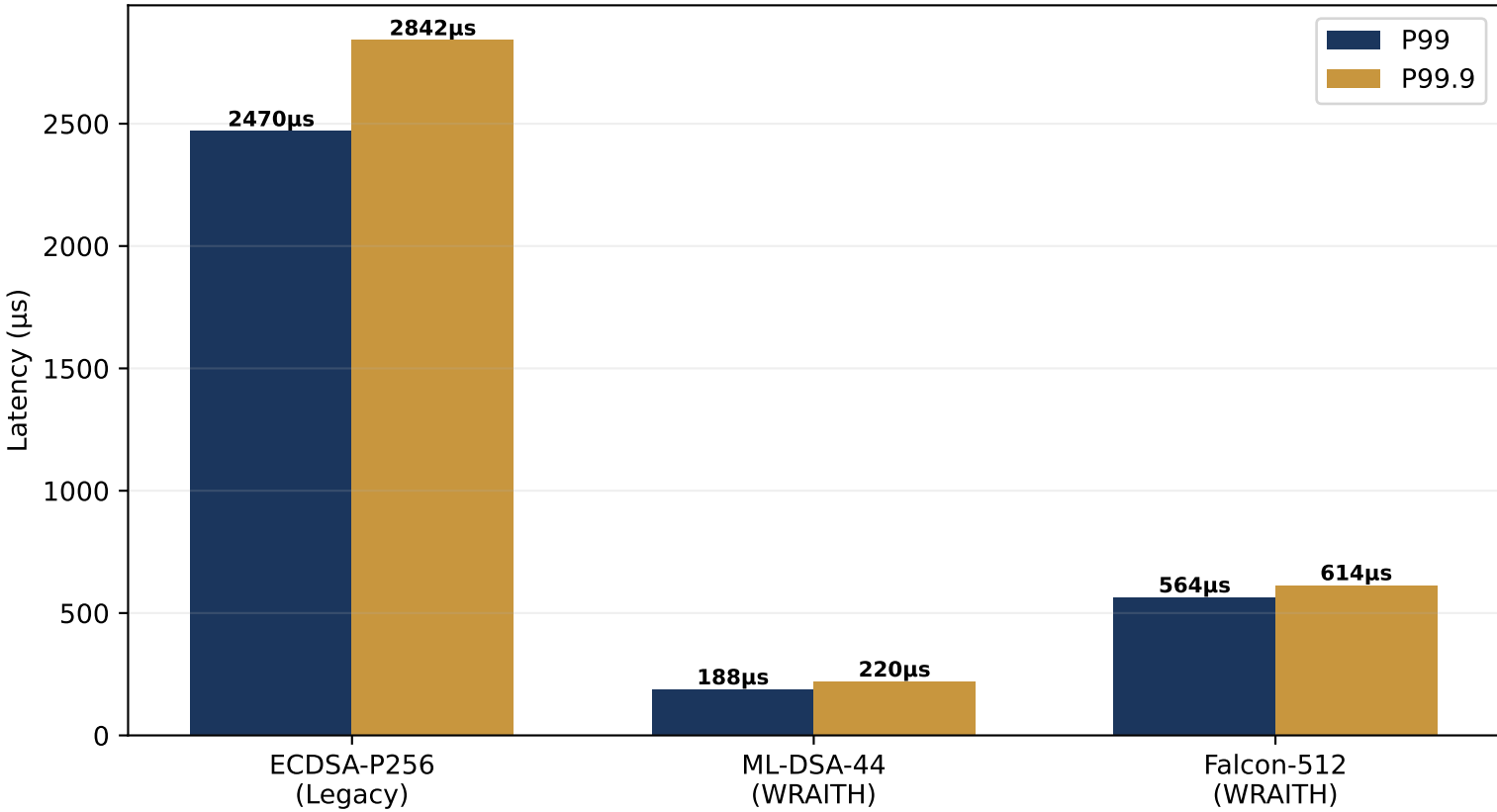
Signing Latency Distribution



Inter-Sample Jitter (Stability)



Tail Latency — P99 & P99.9



SCADA REAL-TIME COMPLIANCE MATRIX

ECDSA-P256 (Legacy):

Mean: 1351μs | P99: 2470μs
Jitter: 406μs | CV: 35.3%
SCADA Safe: ❌ (exceeds 1ms P99)

ML-DSA-44 (WRAITH Injected):

Mean: 101μs | P99: 188μs
Jitter: 34μs | CV: 38.6%
SCADA Safe: ✅ (well within 1ms)
Downtime: 0.000s | Injection: 42ms

Falcon-512 (WRAITH Compact):

Mean: 406μs | P99: 564μs
Jitter: 56μs | CV: 16.2%
SCADA Safe: ❌ (most stable)

VALIDATED FOR:

- Modbus/TCP (10ms cycle)
- DNP3 (100ms poll)
- OPC-UA (1ms publish)
- IEC 61850 GOOSE (4ms)